

Y2 Electromagnetism 2

Non-assessed problem answers 4 (weeks 7–8)

EM waves – Poynting vector – Polarisation



1b) One of the possibilities is

$$\begin{aligned}E_x''(z, t) &= E_0 e^{i(\omega t + kz)}; \\B_y''(z, t) &= -B_0 e^{i(\omega t + kz)}.\end{aligned}$$

1c) The sum of the two waves is

$$\begin{aligned}E_x(z, t) &= 2E_0 e^{i\omega t} \cos(kz); \\B_y(z, t) &= 2B_0 e^{i(\omega t - \pi/2)} \sin(kz).\end{aligned}$$

1d) The amplitudes are $2E_0$ and $2B_0$.

1e) The nodes of the E -field are displaced with respect to the nodes of the B -field by half of the distance between the nodes of each field, i.e. $\Delta z = \pi/2k$. Oscillations of the E -field and the B -field are out of phase by $\pi/2$.

1f) There is no energy flow, $\vec{N} = (\vec{E} \times \vec{B})/\mu_0 = 0$ at the nodes: $z = n\pi/2k$, $n \in \mathbb{Z}$.

2) See lecture 14.

3)

$$E_0 = \sqrt{2\mu_0 c \langle N \rangle} = 194 \text{ V/m}; \quad B_0 = E_0/c = 6.5 \times 10^{-7} \text{ T}.$$

4) The minimal possible thickness is

$$d = \frac{\lambda}{2(n_1 - n_2)} = 500 \text{ } \mu\text{m}.$$